FinShield AI: Intelligent Financial Crime Detection and Investigation Platform

Project Proposal for UCS503P
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1 Higher-order Goal

The higher-order goal of FinShield AI is to make financial crime monitoring faster and more useful. The system will process large numbers of transactions and help identify possible fraud and money-laundering activity. It will give investigators a risk score, the main reasons for the alert, and a clear place to review the case. The project will combine machine learning, simple rules, user behavior analysis, explainable AI, and a case-management workflow. The system will support human investigators rather than making a final legal decision on its own.

2 Time-to-Value

The project will be developed in small steps so that a working system is available early:

- First build transaction upload, basic risk scoring, and the main dashboard.
- Add the fraud detection model using the PaySim dataset.
- Add AML and suspicious-activity analysis using the IBM AMLSim dataset.
- Add rules, behavior analysis, explainable AI, and hybrid risk scoring.
- Add the investigation portal, AI assistant, notifications, and transaction simulation.
- Test and connect all modules during the final weeks.

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3 Problem Statement

Financial crime can appear as a single unusual transaction or as a pattern spread across many transactions and accounts. Fraud and money laundering can be difficult to identify when only one transaction is checked at a time. A simple rule-based system may create many false alerts. A machine-learning model may give a score but may not explain the result. Investigators also need transaction details, related accounts, rules, risk information, and case notes in one place. FinShield AI aims to solve these problems through one web-based platform:

- **Real-Time Transaction Fraud Detection:** Detect possible transaction fraud across incoming payments and transfers.
- **AML Pattern Recognition:** Identify complex, multi-account money laundering schemes and suspicious fund flows.
- **Customer Behavioral Profiling:** Study unusual user activity and flag anomalous deviations from historical baselines.
- **Hybrid Risk Scoring:** Combine distinct rule-based flags and machine learning signals into a unified risk score.
- **Explainable AI (XAI) Insights:** Provide interpretable, feature-level rationales for critical machine learning predictions.

Why this matters now (contextual relevance): With digital payment volumes surging, legacy monitoring creates high operational bottlenecks. Deploying an explainable, hybrid AI workflow directly mitigates alert overload, speeds up case resolution, and enables investigators to trace sophisticated, multi-account laundering rings in real time.

4 Proposed Solution

4.1 Overview

FinShield AI will provide a web-based platform with three major layers: transaction data ingestion, AI/ML-based risk analysis, an alert and case-management workflow. The platform will not automatically declare a transaction criminal. Instead, it will identify risk and provide evidence that a human investigator can review.

4.2 Datasets

1. **PaySim:** used mainly for transaction-level fraud detection and ML model training.
2. **IBM AML:** used mainly for money-laundering and suspicious transaction analysis, including rules, account behavior, and transaction relationships.

The two datasets will not be treated as two separate projects. Both detection parts will send their results to the same risk, alert, and investigation system.

4.3 Main Modules

1. Authentication and role-based access control (RBAC)
2. ML fraud detection with multiple models
3. AML and suspicious-activity analysis

4. Customer/account behavior analysis
5. Hybrid risk scoring
6. Explainable AI using SHAP
7. Fraud and AML investigation portal
8. Admin and analytics dashboard
9. Notification API for email or in-app alerts
10. AI investigation assistant using an LLM API

4.4 Core Workflow

1. Transaction data is uploaded or received by the backend.
2. The system checks the data and prepares the required features.
3. The fraud ML model and AML/rule analysis check the transaction and related activity.
4. Behavior analysis adds information about unusual activity.
5. The hybrid risk engine combines the available scores into a final risk score.
6. High-risk activity creates an alert.
7. The investigator opens the alert and reviews the transaction, rules, behavior, model explanation, and related accounts.
8. A case can be created, assigned, updated, and closed.
9. The AI assistant can prepare a short case summary and suggest useful investigation steps.
10. The final investigator action and important system actions are saved in the audit log.

The IBM AMLSim data will mainly support AML rules, behavior analysis, and account/transaction relationship analysis. The team will keep this part lighter than the main fraud ML pipeline so that the complete project can be finished within three months.

5 Solution Approach

5.1 AI/ML Detection Layer

The main ML work will use the PaySim dataset.

- Clean and prepare transaction data and extarct useful features from amount, transaction type, balances, time, and account activity.
- Compare models such as Logistic Regression, Random Forest, XGBoost, and LightGBM or CatBoost.
- Handle the class imbalance found in fraud data.
- Select the best model using precision, recall, F1-score, ROC-AUC, PR-AUC, and false-positive rate.
- Use SHAP to show which features affected an important prediction.

The IBM AMLSim data will mainly support AML rules, behavior analysis, and account/transaction relationship analysis. The team will keep this part lighter than the main fraud ML pipeline so that the complete project can be finished within the given timeline.

5.2 Rule and Behavior Layer

The rule engine will contain clear rules that can be changed by an authorized administrator. Examples include:

- unusually high transaction amount
- unusually high number of transactions in a short time,transitions, assignment logic
- repeated transfers between related accounts
- sudden change from normal account behavior, and
- suspicious transaction relationships.

Behavior analysis will compare current activity with the account's earlier activity where the required data is available.

5.3 Hybrid Risk Scoring

The final risk score will use several signals instead of relying on one model.

Final Risk Score = f(ML Score, Rule Score, Behavior Score, AML/Network Score)

The exact weights will be tested and documented. The score will be used to prioritize alerts, not to declare that a person or transaction is legally involved in crime.

5.4 Web Architecture

- **Frontend:** React and TypeScript for login, dashboards, transactions, alerts, and investigation pages.
- **Backend API:** FastAPI for authentication, transaction handling, risk analysis, alerts, cases, and reports.
- **ML service:** Python-based service for model loading and prediction.
- **Database:** PostgreSQL for users, transactions, alerts, risk scores, cases, notes, and audit logs.
- **AI service:** one LLM API for investigation summaries and support.
- **Visualization:** charts and simple account/transaction relationship graphs.

6 Evaluation Criteria

The project will be evaluated using both ML and software-system measures.

6.1 Primary Evaluation Metrics

- **Precision:** how many flagged transactions are actually suspicious in the test data.
- **Recall:** how many suspicious transactions are detected.
- **F1-score:** a balance between precision and recall.
- **ROC-AUC and PR-AUC:** overall model performance, with PR-AUC being useful for imbalanced fraud data.

- **False-positive rate:** how many normal transactions are wrongly flagged.
- **Response time:** time needed to return a risk result.

6.2 Secondary Evaluation Metrics

- Alert prioritization quality
- Case workflow completion
- Dashboard usability
- API reliability
- Successful transaction-to-alert-to-case processing

7 Scalability

The project is an academic prototype, but its design will allow larger data volumes later.

1. Use database indexes and pagination for common queries.
2. Support batch processing for large transaction files.
3. Keep the ML prediction part separate from the main application logic.
4. Keep model and preprocessing versions clear so a model can be replaced without changing the complete application.
5. Use Docker so the main services can be moved to another machine or cloud platform later.

8 Engine Availability Heuristic

A mature set of "engines" (tooling/services) is available to implement this as a web-based project:

- Python, pandas, NumPy, scikit-learn, XGBoost, and SHAP for ML.
- FastAPI and Pydantic for the backend.
- React and TypeScript for the frontend.
- PostgreSQL for data storage.
- Docker and Docker Compose for setup and deployment.
- A single LLM API for the AI investigation assistant.
- GitHub for version control and team collaboration.

We will use these mature components to focus on workflow design, correctness, and measurement rather than inventing new infrastructure.

9 Project Scope and Deliverables

9.1 Initial Deliverables

- Working web interface with authentication and RBAC.
- Transaction upload and management.
- PaySim-based fraud ML pipeline with model comparison.
- IBM AMLSim-based AML and suspicious-activity analysis.
- Configurable rule engine and Behavior analysis.
- Hybrid risk scoring.
- SHAP-based explanation for ML predictions.

9.2 Subsequent Deliverables

- Alert dashboard with filtering and risk priority.
- Investigation and case-management workflow.
- Basic account/transaction relationship graph.
- AI investigation assistant.
- Real-time transaction simulation.
- Email or in-app notifications.
- Audit logs.
- Docker-based project setup and documentation.

10 Risks and Mitigations

- **Dataset limitations:** Clearly document dataset limitations and use PaySim and IBM AMLSim for their respective purposes.
- **False alerts and ML errors:** Use multiple evaluation metrics, rule-based checks, and SHAP explanations to improve reliability.
- **Integration and time constraints:** Define APIs early, integrate modules regularly, and complete core features before advanced features.
- **Security and API risks:** Use authentication, RBAC, input validation, and keep the AI assistant as a supporting feature.
- **Incorrect decisions:** The system will provide risk indicators only; final decisions will remain with the human investigator.

11 Summary

FinShield AI aims to provide one platform for detecting and investigating financial crime. The project will combine PaySim-based fraud detection with IBM AMLSim-based moneylaundering and suspicious-activity analysis. Rules, behavior analysis, hybrid risk scoring, SHAP, case management, dashboards, and an AI investigation assistant will bring these parts together. The project is planned for a four-member team over three months. The system will focus on useful features, clear software design, measurable ML results, testing, and a simple investigation workflow. This approach keeps the project realistic while giving it strong value for both Software Engineering and Machine Learning work.